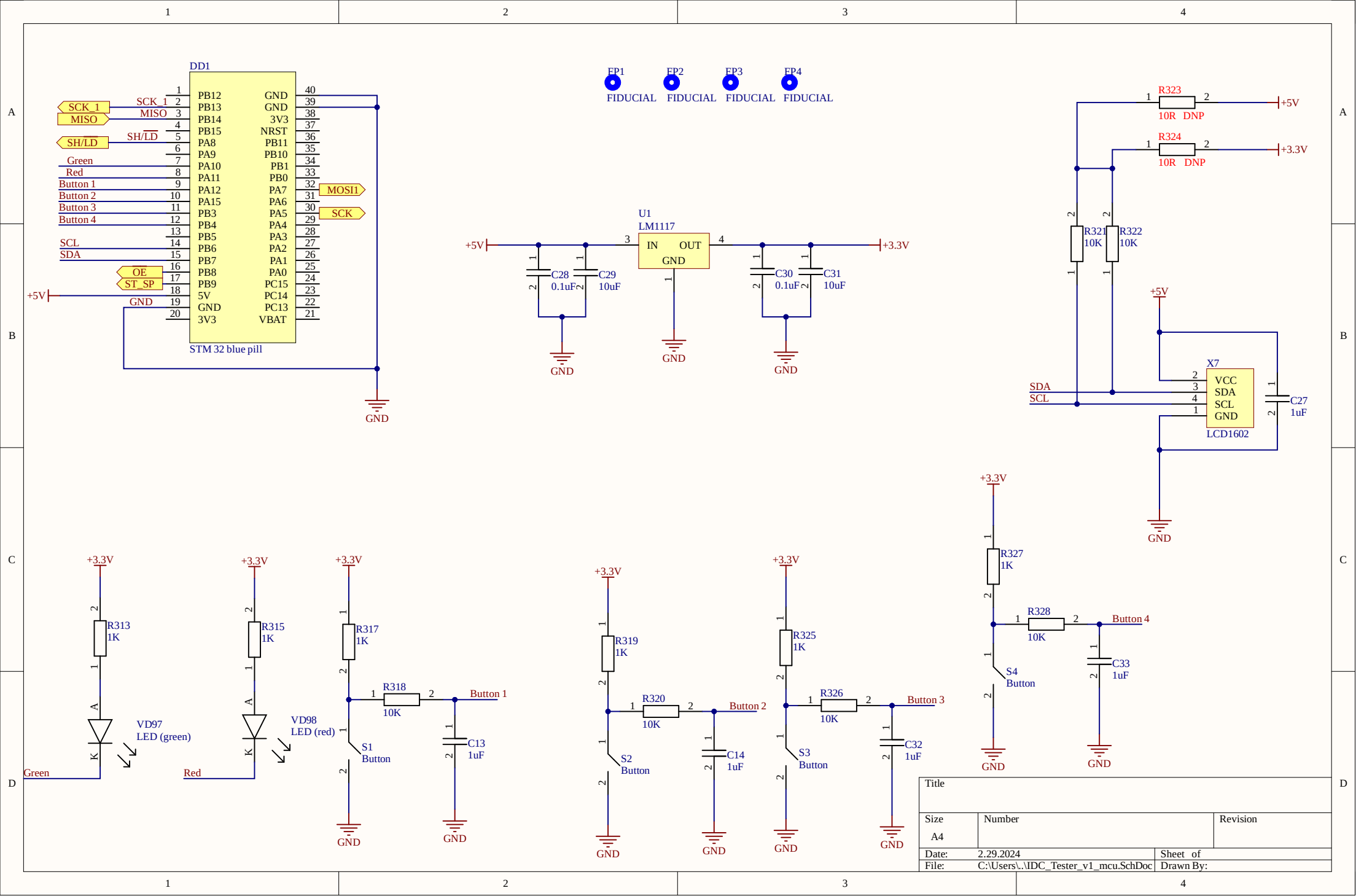

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PB12
PB13
PB14
PB15
PA8
PA9
PA10
PA11
PA12
PA15
PB3
PB4
PB5
PB6
PB7
PB8
PB9
5V
GND
3V3
GND
GND
3V3
NRST
PB11
PB10
PB1
PB0
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
PC15
PC14
PC13
VBAT
SCK_1
MISO
SH/LD
Green
Red
Button 1
Button 2
Button 3
Button 4
SCL
SDA
OE
ST_SP
MOSI1
SCK
STM 32 blue pill
FP1
FP2
FP3
FP4
FIDUCIAL
FIDUCIAL
FIDUCIAL
FIDUCIAL
U1
LM1117
IN
OUT
GND
+5V
+3.3V
C28
0.1uF
C29
10uF
C30
0.1uF
C31
10uF
R323
10R DNP
R324
10R DNP
R321
10K
R322
10K
X7
VCC
SDA
SCL
GND
LCD1602
C27
1uF
R313
1K
VD97
LED (green)
R315
1K
VD98
LED (red)
R317
1K
R318
10K
S1
Button
Button 1
C13
1uF
R319
1K
R320
10K
S2
Button
Button 2
C14
1uF
R325
1K
R326
10K
S3
Button
Button 3
C32
1uF
R327
1K
R328
10K
S4
Button
Button 4
C33
1uF
Title
Size
A4
Number
Revision
Date:
2.29.2024
Sheet of
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Drawn By:

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REVISION	DESCRIPTION	DATE	APPROVED

DD2 74HC595D,118 — +3.3V, GND, C1 1uF, SCK, ST_SP, MOSI1, OE; Q0–Q7 → Port 1–Port 8 via R217–R224 100R; Q7S → MOSI 2

DD3 74HC595D,118 — +3.3V, GND, C2 1uF, SCK, ST_CP, MOSI 2, OE; Q0–Q7 → Port 9–Port 16 via R225–R232 100R; Q7S → MOSI 3

DD4 74HC595D,118 — +3.3V, GND, C3 1uF, SCK, ST_CP, MOSI 3, OE; Q0–Q7 → Port 17–Port 24 via R233–R240 100R; Q7S → MOSI 4

DD5 74HC595D,118 — +3.3V, GND, C4 1uF, SCK, ST_CP, MOSI 4, OE; Q0–Q7 → Port 25–Port 32 via R241–R248 100R; Q7S → MOSI 5

DD6 74HC595D,118 — +3.3V, GND, C5 1uF, SCK, ST_CP, MOSI 5, OE; Q0–Q7 → Port 33–Port 40 via R249–R256 100R; Q7S → MOSI 6

DD7 74HC595D,118 — +3.3V, GND, C6 1uF, SCK, ST_CP, MOSI 6, OE; Q0–Q7 → Port 41–Port 48 via R257–R264 100R; Q7S → MOSI 7

DD8 74HC595D,118 — +3.3V, GND, C7 1uF, SCK, ST_CP, MOSI 7, OE; Q0–Q7 → Port 49–Port 56 via R265–R272 100R; Q7S → MOSI 8

DD9 74HC595D,118 — +3.3V, GND, C8 1uF, SCK, ST_CP, MOSI 8, OE; Q0–Q7 → Port 57–Port 64 via R273–R280 100R; Q7S → MOSI 9

DD10 74HC595D,118 — +3.3V, GND, C9 1uF, SCK, ST_CP, MOSI 9, OE; Q0–Q7 → Port 65–Port 72 via R281–R288 100R; Q7S → MOSI 10

DD11 74HC595D,118 — +3.3V, GND, C10 1uF, SCK, ST_CP, MOSI 10, OE; Q0–Q7 → Port 73–Port 80 via R289–R296 100R; Q7S → MOSI 11

DD12 74HC595D,118 — +3.3V, GND, C11 1uF, SCK, ST_CP, MOSI 11, OE; Q0–Q7 → Port 81–Port 88 via R297–R304 100R; Q7S → MOSI 12

DD13 74HC595D,118 — +3.3V, GND, C12 1uF, SCK, ST_CP, MOSI 12, OE; Q0–Q7 → Port 89–Port 96 via R305–R312 100R; Q7S → 9

OE → R205–R216 10K → GND

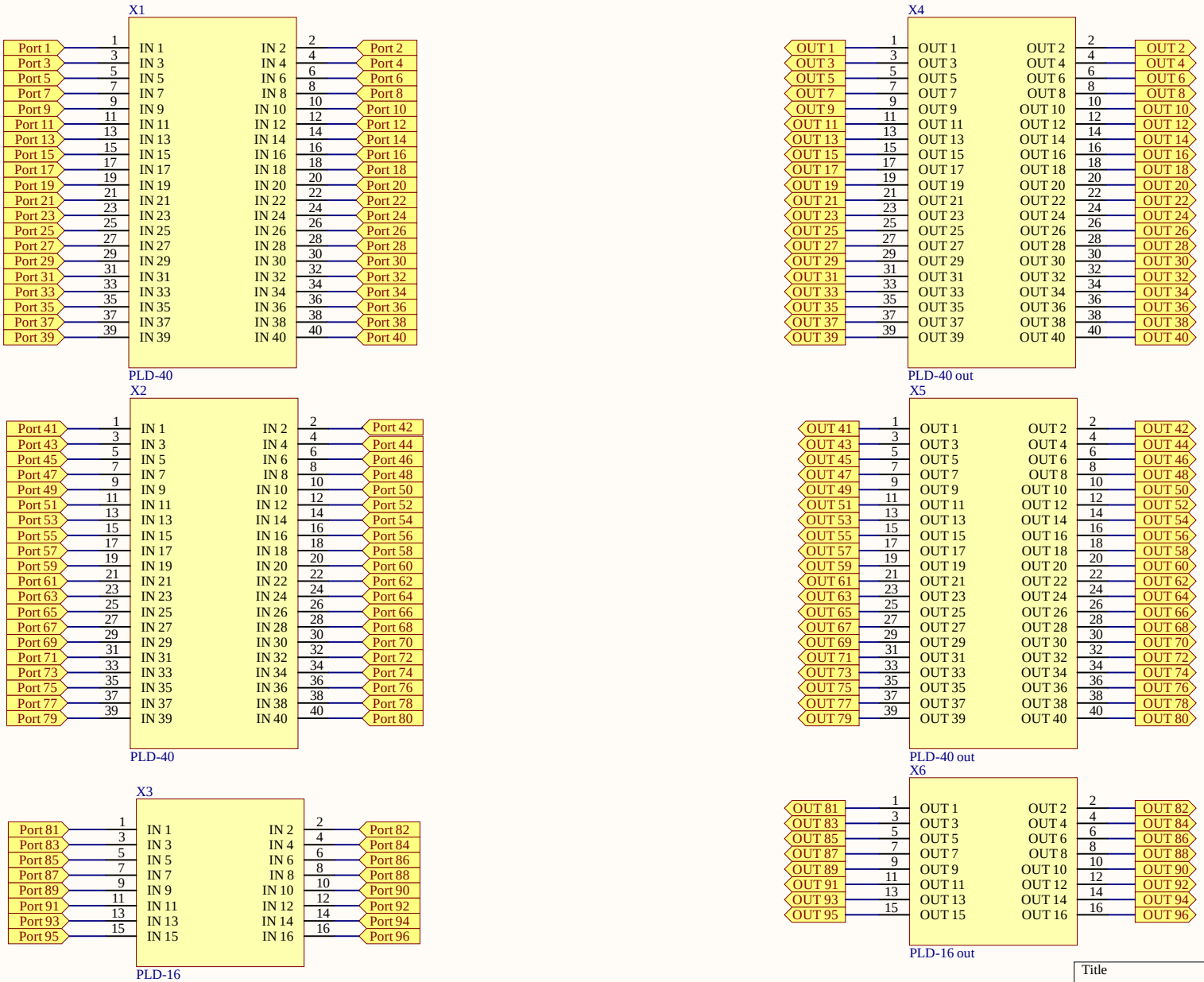
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DSN: *	
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PROJECT | PROJECT REVISION: | DOCUMENT REVISION: | DESIGN ITEM ID: 8d0e6c0d72c69ad79fd5769b6ebe49aa4

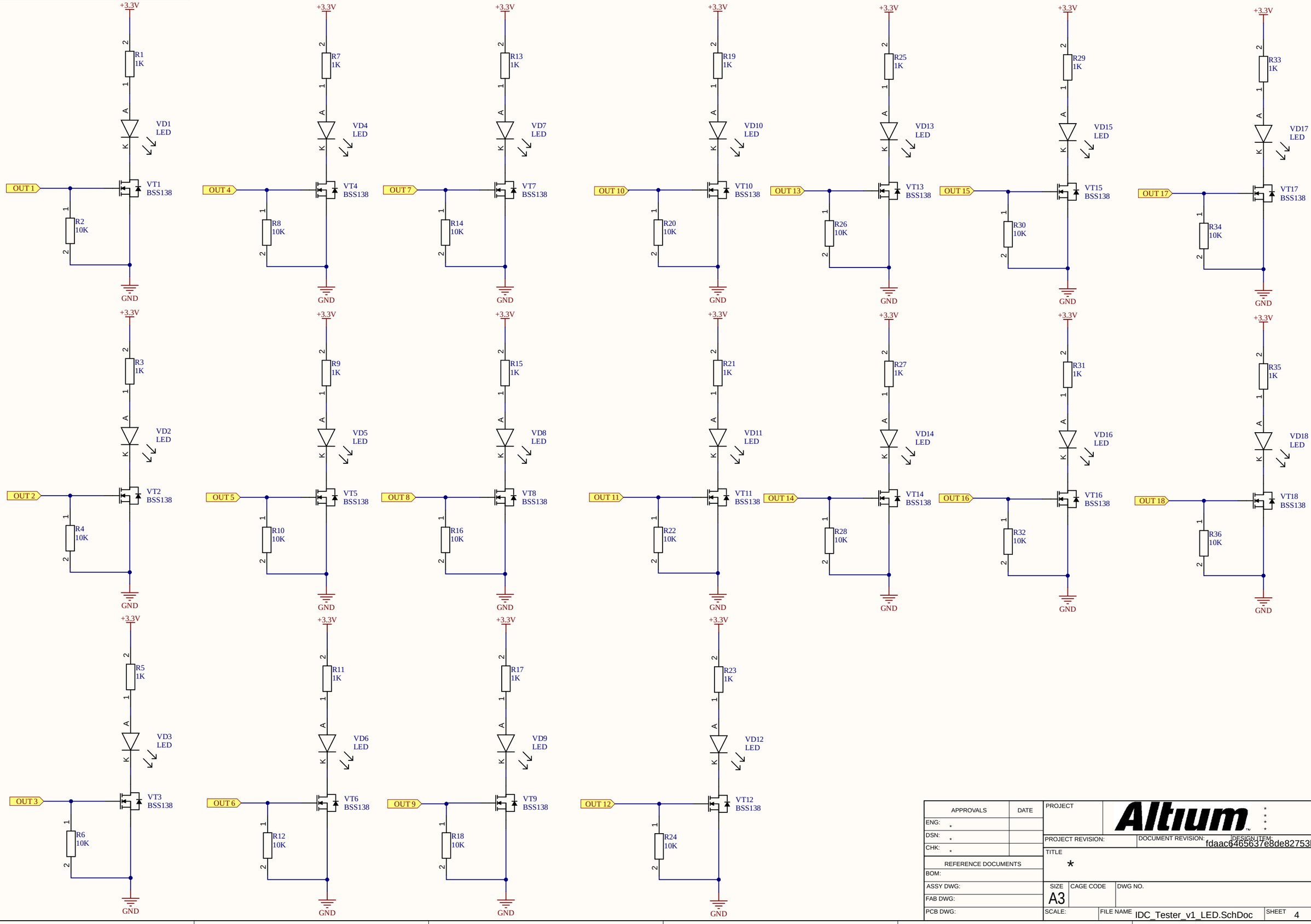
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X1
Port 1 1 IN 1 IN 2 2 Port 2
Port 3 3 IN 3 IN 4 4 Port 4
Port 5 5 IN 5 IN 6 6 Port 6
Port 7 7 IN 7 IN 8 8 Port 8
Port 9 9 IN 9 IN 10 10 Port 10
Port 11 11 IN 11 IN 12 12 Port 12
Port 13 13 IN 13 IN 14 14 Port 14
Port 15 15 IN 15 IN 16 16 Port 16
Port 17 17 IN 17 IN 18 18 Port 18
Port 19 19 IN 19 IN 20 20 Port 20
Port 21 21 IN 21 IN 22 22 Port 22
Port 23 23 IN 23 IN 24 24 Port 24
Port 25 25 IN 25 IN 26 26 Port 26
Port 27 27 IN 27 IN 28 28 Port 28
Port 29 29 IN 29 IN 30 30 Port 30
Port 31 31 IN 31 IN 32 32 Port 32
Port 33 33 IN 33 IN 34 34 Port 34
Port 35 35 IN 35 IN 36 36 Port 36
Port 37 37 IN 37 IN 38 38 Port 38
Port 39 39 IN 39 IN 40 40 Port 40
PLD-40
X2
Port 41 1 IN 1 IN 2 2 Port 42
Port 43 3 IN 3 IN 4 4 Port 44
Port 45 5 IN 5 IN 6 6 Port 46
Port 47 7 IN 7 IN 8 8 Port 48
Port 49 9 IN 9 IN 10 10 Port 50
Port 51 11 IN 11 IN 12 12 Port 52
Port 53 13 IN 13 IN 14 14 Port 54
Port 55 15 IN 15 IN 16 16 Port 56
Port 57 17 IN 17 IN 18 18 Port 58
Port 59 19 IN 19 IN 20 20 Port 60
Port 61 21 IN 21 IN 22 22 Port 62
Port 63 23 IN 23 IN 24 24 Port 64
Port 65 25 IN 25 IN 26 26 Port 66
Port 67 27 IN 27 IN 28 28 Port 68
Port 69 29 IN 29 IN 30 30 Port 70
Port 71 31 IN 31 IN 32 32 Port 72
Port 73 33 IN 33 IN 34 34 Port 74
Port 75 35 IN 35 IN 36 36 Port 76
Port 77 37 IN 37 IN 38 38 Port 78
Port 79 39 IN 39 IN 40 40 Port 80
PLD-40
X3
Port 81 1 IN 1 IN 2 2 Port 82
Port 83 3 IN 3 IN 4 4 Port 84
Port 85 5 IN 5 IN 6 6 Port 86
Port 87 7 IN 7 IN 8 8 Port 88
Port 89 9 IN 9 IN 10 10 Port 90
Port 91 11 IN 11 IN 12 12 Port 92
Port 93 13 IN 13 IN 14 14 Port 94
Port 95 15 IN 15 IN 16 16 Port 96
PLD-16
X4
OUT 1 1 OUT 1 OUT 2 2 OUT 2
OUT 3 3 OUT 3 OUT 4 4 OUT 4
OUT 5 5 OUT 5 OUT 6 6 OUT 6
OUT 7 7 OUT 7 OUT 8 8 OUT 8
OUT 9 9 OUT 9 OUT 10 10 OUT 10
OUT 11 11 OUT 11 OUT 12 12 OUT 12
OUT 13 13 OUT 13 OUT 14 14 OUT 14
OUT 15 15 OUT 15 OUT 16 16 OUT 16
OUT 17 17 OUT 17 OUT 18 18 OUT 18
OUT 19 19 OUT 19 OUT 20 20 OUT 20
OUT 21 21 OUT 21 OUT 22 22 OUT 22
OUT 23 23 OUT 23 OUT 24 24 OUT 24
OUT 25 25 OUT 25 OUT 26 26 OUT 26
OUT 27 27 OUT 27 OUT 28 28 OUT 28
OUT 29 29 OUT 29 OUT 30 30 OUT 30
OUT 31 31 OUT 31 OUT 32 32 OUT 32
OUT 33 33 OUT 33 OUT 34 34 OUT 34
OUT 35 35 OUT 35 OUT 36 36 OUT 36
OUT 37 37 OUT 37 OUT 38 38 OUT 38
OUT 39 39 OUT 39 OUT 40 40 OUT 40
PLD-40 out
X5
OUT 41 1 OUT 1 OUT 2 2 OUT 42
OUT 43 3 OUT 3 OUT 4 4 OUT 44
OUT 45 5 OUT 5 OUT 6 6 OUT 46
OUT 47 7 OUT 7 OUT 8 8 OUT 48
OUT 49 9 OUT 9 OUT 10 10 OUT 50
OUT 51 11 OUT 11 OUT 12 12 OUT 52
OUT 53 13 OUT 13 OUT 14 14 OUT 54
OUT 55 15 OUT 15 OUT 16 16 OUT 56
OUT 57 17 OUT 17 OUT 18 18 OUT 58
OUT 59 19 OUT 19 OUT 20 20 OUT 60
OUT 61 21 OUT 21 OUT 22 22 OUT 62
OUT 63 23 OUT 23 OUT 24 24 OUT 64
OUT 65 25 OUT 25 OUT 26 26 OUT 66
OUT 67 27 OUT 27 OUT 28 28 OUT 68
OUT 69 29 OUT 29 OUT 30 30 OUT 70
OUT 71 31 OUT 31 OUT 32 32 OUT 72
OUT 73 33 OUT 33 OUT 34 34 OUT 74
OUT 75 35 OUT 35 OUT 36 36 OUT 76
OUT 77 37 OUT 37 OUT 38 38 OUT 78
OUT 79 39 OUT 39 OUT 40 40 OUT 80
PLD-40 out
X6
OUT 81 1 OUT 1 OUT 2 2 OUT 82
OUT 83 3 OUT 3 OUT 4 4 OUT 84
OUT 85 5 OUT 5 OUT 6 6 OUT 86
OUT 87 7 OUT 7 OUT 8 8 OUT 88
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OUT 95 15 OUT 15 OUT 16 16 OUT 96
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Size
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Number
Revision
Date: 2.29.2024
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REVISION	DESCRIPTION	DATE	APPROVED



APPROVALS	DATE
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FAB DWG:	
PCB DWG:	

PROJECT: Altium *

PROJECT REVISION: | DOCUMENT REVISION: | IDESIGN ITEM: fdaac6465637e8de82753b878e8013cf9

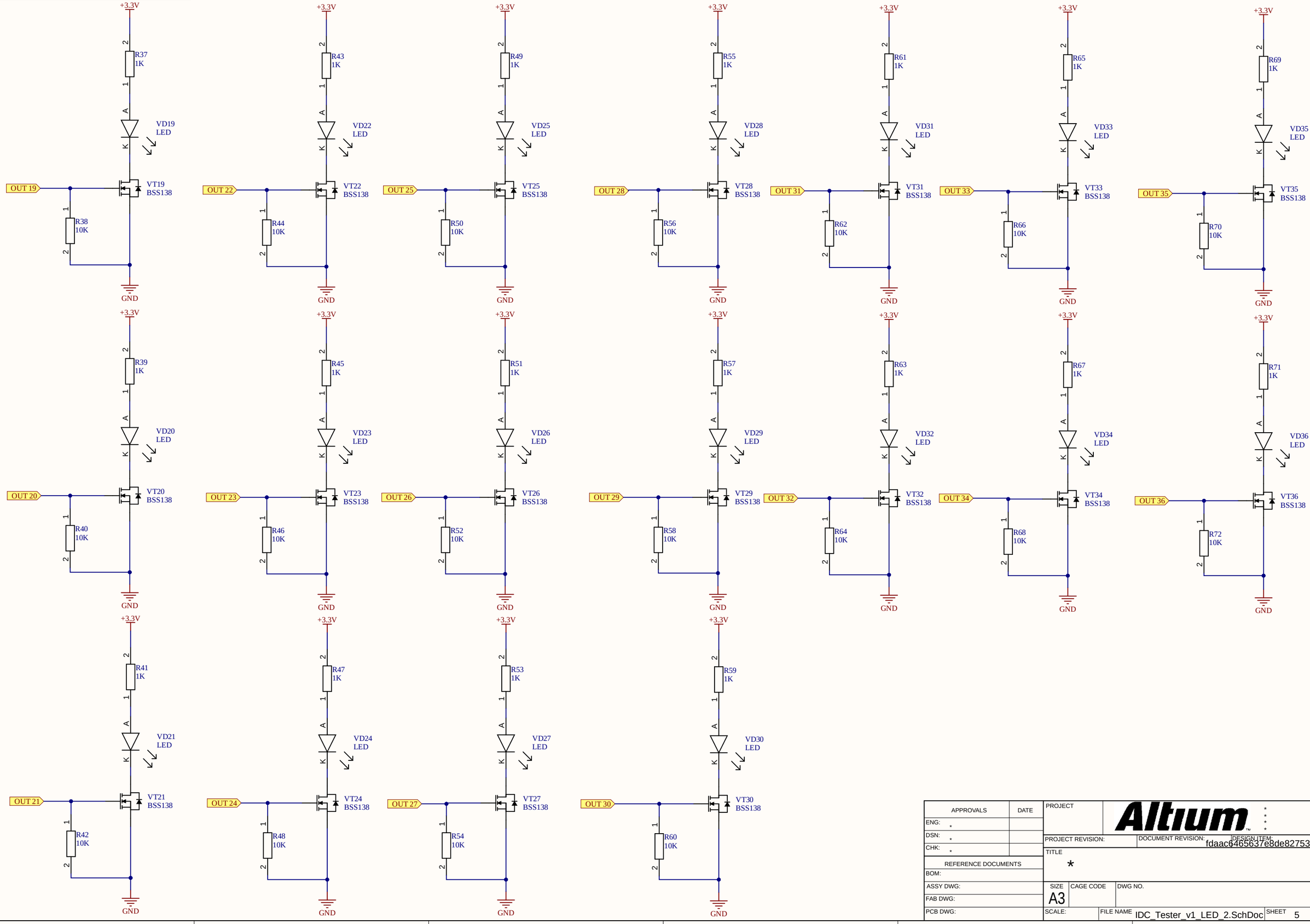
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APPROVALS	DATE
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REFERENCE DOCUMENTS	
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ASSY DWG:	
FAB DWG:	
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PROJECT

Altium

PROJECT REVISION:

DOCUMENT REVISION:

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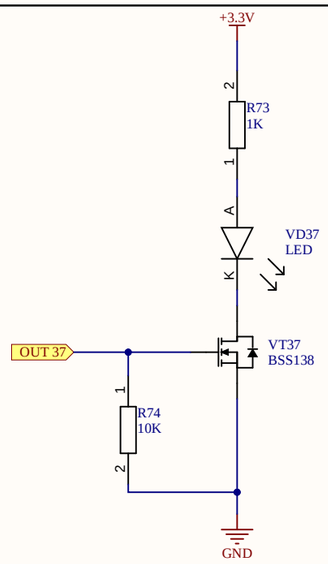
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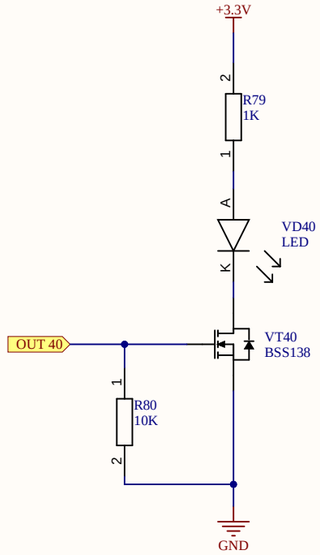
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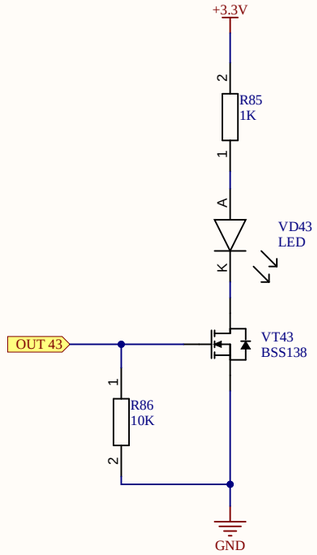
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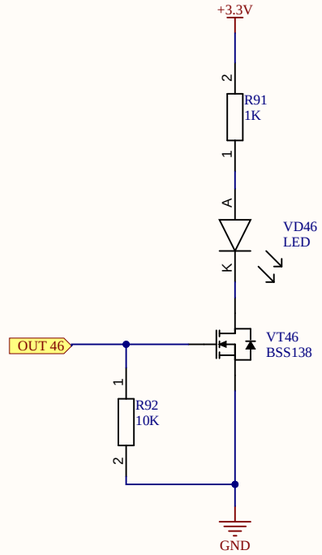
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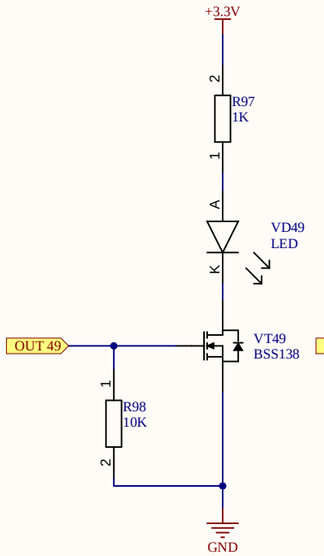
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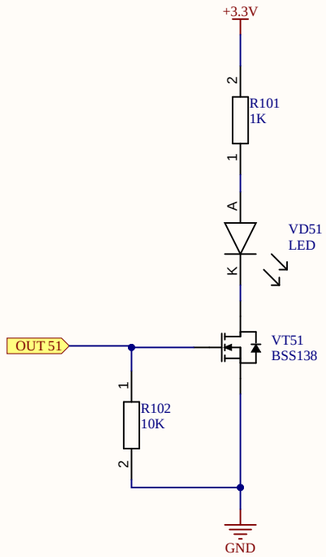


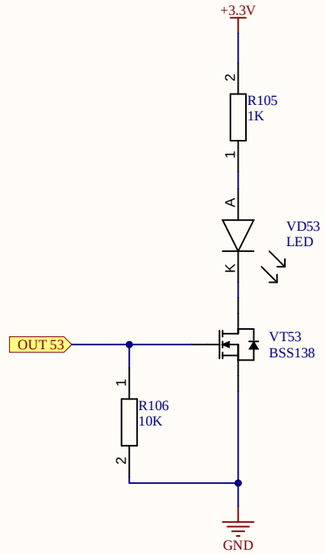


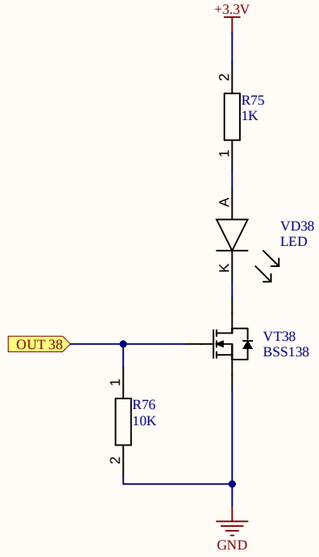


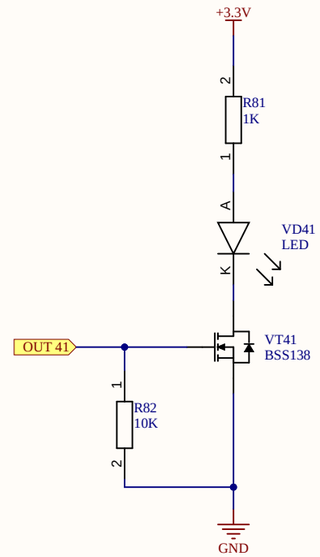


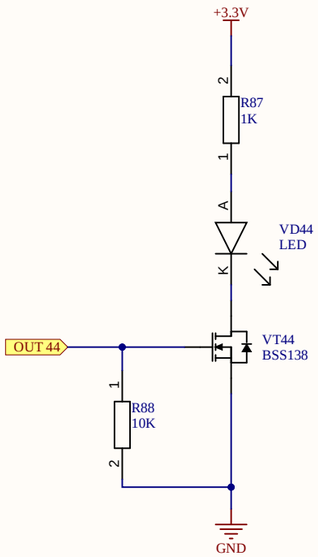


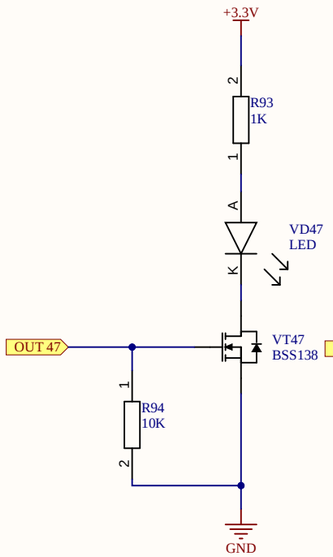


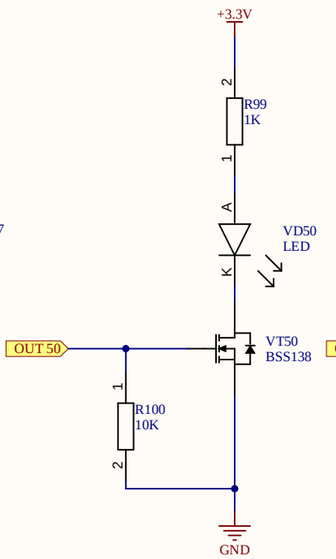


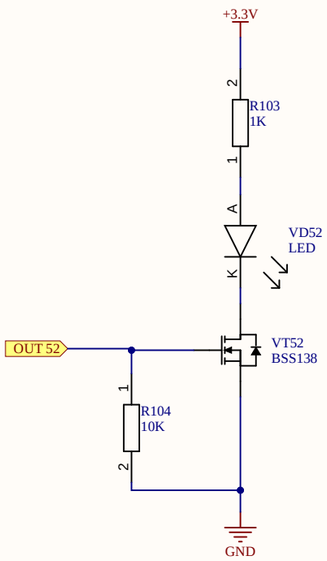


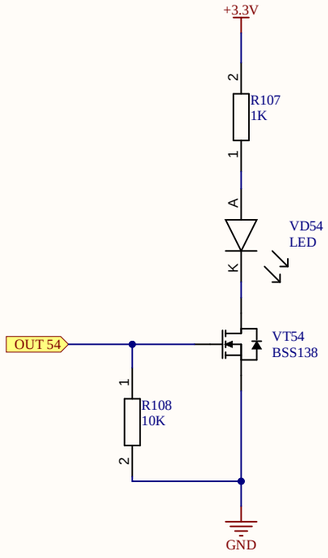


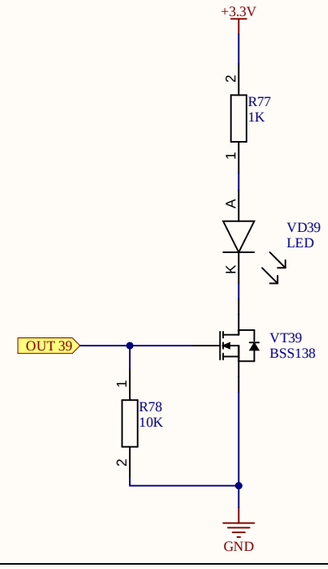


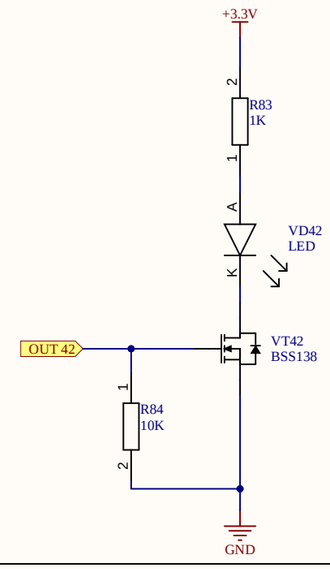


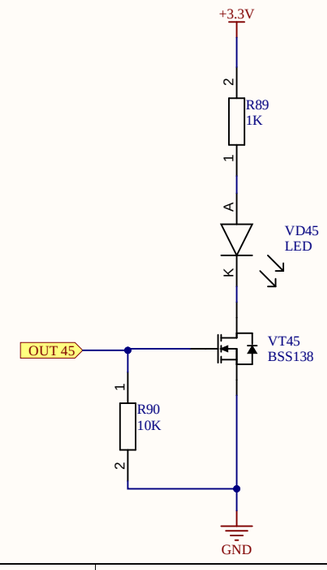


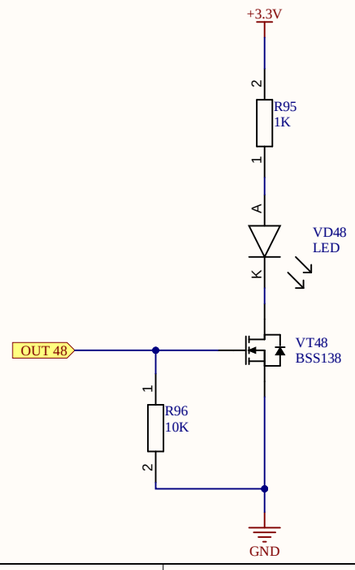












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PROJECT

Altium

PROJECT REVISION:

DOCUMENT REVISION:

IDESIGN ITEM: 63b5c91b652582850aa473f555385db6

TITLE *

SIZE	CAGE CODE	DWG NO.	REV
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SCALE: | FILE NAME IDC_Tester_v1_LED_3.SchDoc | SHEET 6 OF 10

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REVISION	DESCRIPTION	DATE	APPROVED

+3.3V
R109 1K
VD55 LED
OUT 55
VT55 BSS138
R110 10K
GND

+3.3V
R115 1K
VD58 LED
OUT 58
VT58 BSS138
R116 10K
GND

+3.3V
R121 1K
VD61 LED
OUT 61
VT61 BSS138
R122 10K
GND

+3.3V
R127 1K
VD64 LED
OUT 64
VT64 BSS138
R128 10K
GND

+3.3V
R133 1K
VD67 LED
OUT 67
VT67 BSS138
R134 10K
GND

+3.3V
R137 1K
VD69 LED
OUT 69
VT69 BSS138
R138 10K
GND

+3.3V
R141 1K
VD71 LED
OUT 71
VT71 BSS138
R142 10K
GND

+3.3V
R111 1K
VD56 LED
OUT 56
VT56 BSS138
R112 10K
GND

+3.3V
R117 1K
VD59 LED
OUT 59
VT59 BSS138
R118 10K
GND

+3.3V
R123 1K
VD62 LED
OUT 62
VT62 BSS138
R124 10K
GND

+3.3V
R129 1K
VD65 LED
OUT 65
VT65 BSS138
R130 10K
GND

+3.3V
R135 1K
VD68 LED
OUT 68
VT68 BSS138
R136 10K
GND

+3.3V
R139 1K
VD70 LED
OUT 70
VT70 BSS138
R140 10K
GND

+3.3V
R143 1K
VD72 LED
OUT 72
VT72 BSS138
R144 10K
GND

+3.3V
R113 1K
VD57 LED
OUT 57
VT57 BSS138
R114 10K
GND

+3.3V
R119 1K
VD60 LED
OUT 60
VT60 BSS138
R120 10K
GND

+3.3V
R125 1K
VD63 LED
OUT 63
VT63 BSS138
R126 10K
GND

+3.3V
R131 1K
VD66 LED
OUT 66
VT66 BSS138
R132 10K
GND

APPROVALS	DATE
ENG: *	
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REFERENCE DOCUMENTS	
BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

PROJECT: Altium *

PROJECT REVISION: | DOCUMENT REVISION: | DESIGN ITEM: 17c22b0c3be1e018a709ca5d1cf23a176

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REVISION	DESCRIPTION	DATE	APPROVED

+3.3V
R145 1K
VD73 LED
OUT 73
VT73 BSS138
R146 10K
GND

+3.3V
R151 1K
VD76 LED
OUT 76
VT76 BSS138
R152 10K
GND

+3.3V
R157 1K
VD79 LED
OUT 79
VT79 BSS138
R158 10K
GND

+3.3V
R163 1K
VD82 LED
OUT 82
VT82 BSS138
R164 10K
GND

+3.3V
R169 1K
VD85 LED
OUT 85
VT85 BSS138
R170 10K
GND

+3.3V
R173 1K
VD87 LED
OUT 87
VT87 BSS138
R174 10K
GND

+3.3V
R177 1K
VD89 LED
OUT 89
VT89 BSS138
R178 10K
GND

+3.3V
R147 1K
VD74 LED
OUT 74
VT74 BSS138
R148 10K
GND

+3.3V
R153 1K
VD77 LED
OUT 77
VT77 BSS138
R154 10K
GND

+3.3V
R159 1K
VD80 LED
OUT 80
VT80 BSS138
R160 10K
GND

+3.3V
R165 1K
VD83 LED
OUT 83
VT83 BSS138
R166 10K
GND

+3.3V
R171 1K
VD86 LED
OUT 86
VT86 BSS138
R172 10K
GND

+3.3V
R175 1K
VD88 LED
OUT 88
VT88 BSS138
R176 10K
GND

+3.3V
R179 1K
VD90 LED
OUT 90
VT90 BSS138
R180 10K
GND

+3.3V
R149 1K
VD75 LED
OUT 75
VT75 BSS138
R150 10K
GND

+3.3V
R155 1K
VD78 LED
OUT 78
VT78 BSS138
R156 10K
GND

+3.3V
R161 1K
VD81 LED
OUT 81
VT81 BSS138
R162 10K
GND

+3.3V
R167 1K
VD84 LED
OUT 84
VT84 BSS138
R168 10K
GND

APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	
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BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

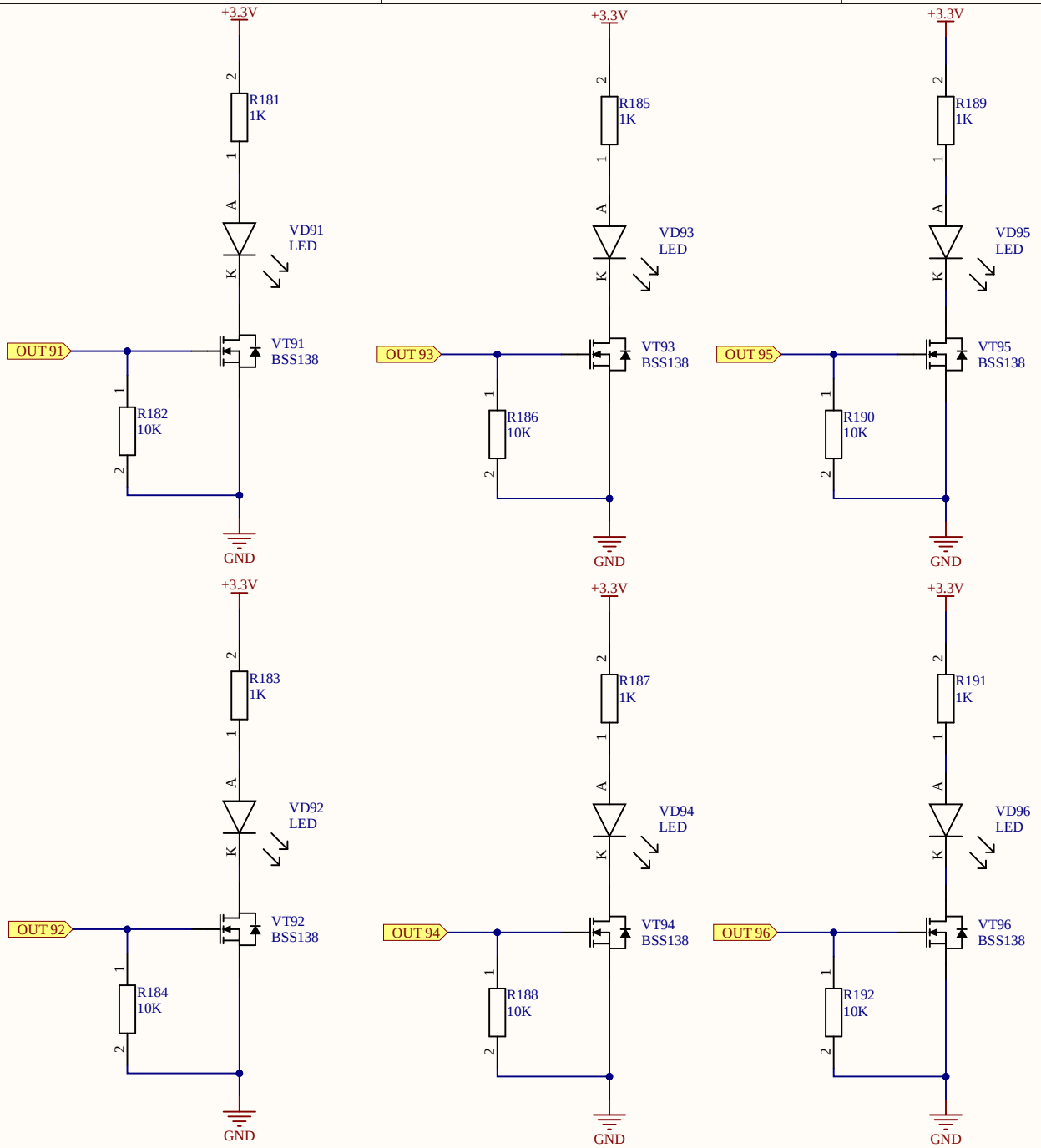
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A3			

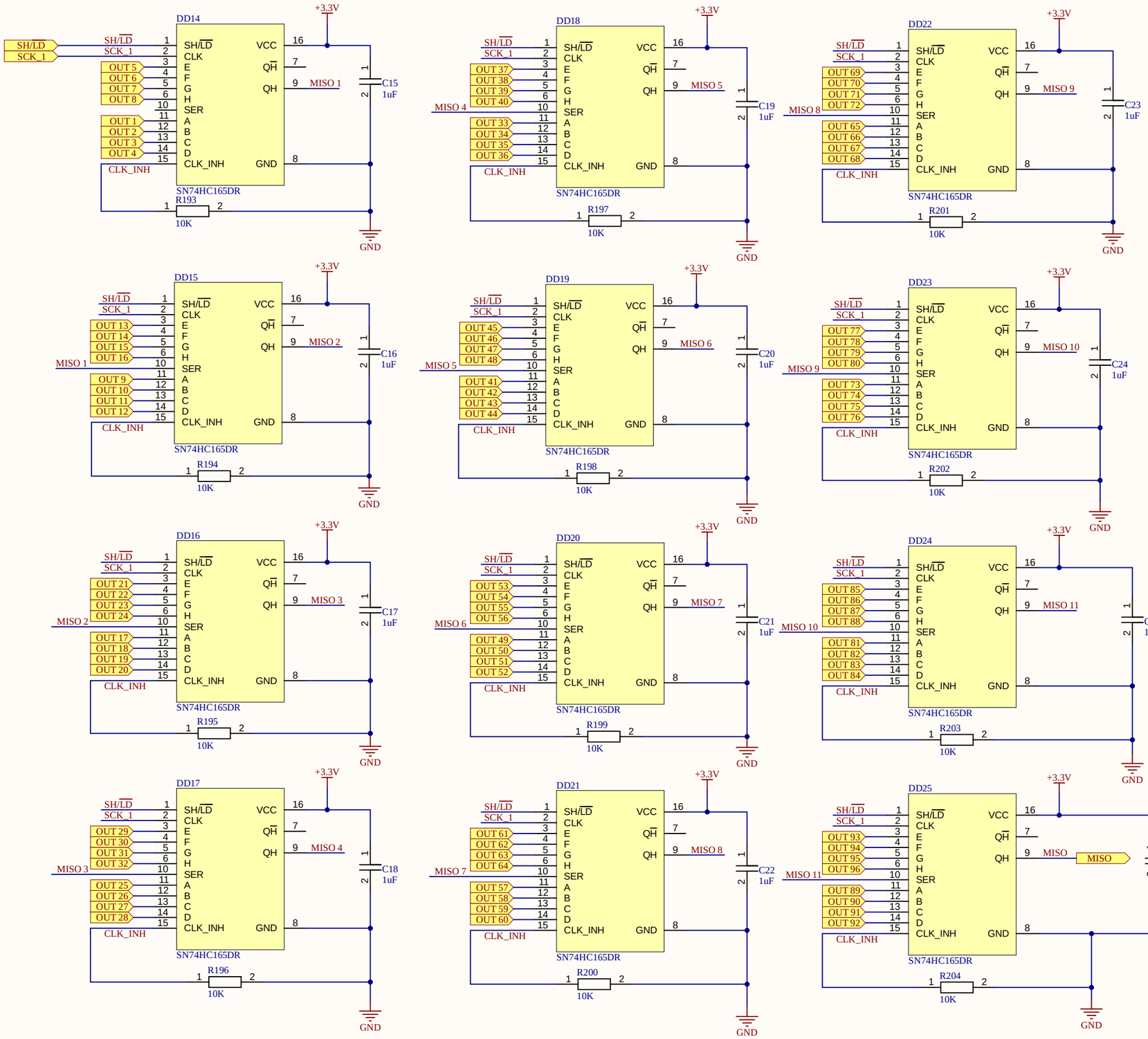
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Title		
Size A4	Number	Revision
Date: 2.29.2024	Sheet of	
File: C:\Users\..\IDC_Tester_v1_LED_6.SchDoc	Drawn By:	

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REVISION	DESCRIPTION	DATE	APPROVED



APPROVALS	DATE
ENG: *	
DSN: *	
CHK: *	

REFERENCE DOCUMENTS

BOM:

ASSY DWG:

FAB DWG:

PCB DWG:

PROJECT

PROJECT REVISION: DOCUMENT REVISION: DESIGN ITEM:

TITLE

*

SIZE	CAGE CODE	DWG NO.	REV
A3			

SCALE: FILE NAME IDC_Tester_v1_sift_register_input.SchDoc SHEET 10 OF 10

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